

Student Name: **Oladipupo Roland Familua**

Student Number: **2326310**

Location/Site: **City Campus**

Module Code:**6CS007**

Module Name: **Project and Professionalism**

Project Title:

Optimizing and Prompting Artificial Intelligence for Enhanced User Interaction in Social Feed Environments

Supervisor: **Jeffrey Ting**

Reader Name: **Hiran Patel**

Submission Date:
(The date you handed in the assessment)

Award Title**: BSc (Hons) Computer Science**
(Award Title for your project, if in doubt refer to your course/Module Registration)

Declaration Sheet

Presented in partial fulfilment of the assessment requirements for the above award.

This work or any part thereof has not previously been presented in any form to the University or to any other institutional body whether for assessment or for other purposes. Save for any express acknowledgements, references and/or bibliographies cited in the work. I confirm that the intellectual contents of the work are the result of my own efforts and of no other person.

It is acknowledged that the author of any project work shall own the copyright. However, by submitting such copyright work for assessment, the author grants to the University a perpetual royalty-free licence to do all or any of those things referred to in section 16(i) of the Copyright Designs and Patents Act 1988. (viz: to copy work; to issue copies to the public; to perform or show or play the work in public; to broadcast the work or to make an adaptation of the work).

Student Name (Print):

Student Number:

Signature: ……..……. Date: ……

(Must include the unedited statement above. Sign and date)

Please use an electronic signature (scan and insert)

Table of Contents **3&4**

1 Chapter 1: Introduction

1.1 Project Motivation .....
1.2 Background of Artificial Intelligence in Social Platforms .....
1.3 Overview of Prompting and Optimization Techniques .....
1.4 Problem Statement .....
1.5 Aims and Objectives .....
  1.5.1 Academic Research Question .....
  1.5.2 Aims .....
  1.5.3 Objectives .....
1.6 Scope and Limitations .....
1.7 Project Relevance and Real-World Impact .....
1.8 Report Framework .....
1.9 Summary .....

---

2 Chapter 2: Literature Review
2.1 Introduction .....
2.2 Overview of AI in Social Media and Feed Environments .....
2.3 AI Prompting Techniques and Conversational Optimization .....
2.4 Machine Learning for Personalization and Recommendation Systems .....
2.5 Enhancing User Interaction through AI-Powered Systems .....
2.6 Ethical and Social Implications of AI in User Interaction .....
2.7 Frameworks and Technologies Supporting AI Optimization .....
2.8 Summary .....

---

3 Chapter 3: System Design and Architecture
3.1 Introduction .....
3.2 Overview of SOULSTRIVENET Application
.....
3.3 Functional and Non-Functional Requirements .....
3.4 System Architecture .....
3.5 Data Flow and AI Interaction Model .....
3.6 Design of the AI Chatbot and Prompting Mechanism .....
3.7 Database and Data Management Architecture .....
3.8 Tools, Frameworks, and Technologies .....
3.9 Summary .....

---

4 Chapter 4: Implementation and Development
4.1 Introduction .....
4.2 Backend Development (Node.JS) .....
4.3 Frontend and User Interface (React Native) .....
4.4 Integration of AI Chatbot and Prompt Engineering .....
    4.4.1 System Prompt Structure
    4.4.2 User Query Pipeline
    4.4.3 Context Injection Logic
    4.4.4 Example of Poor vs Improved Prompts
    4.4.5 Example Output Differences
    4.4.6 Fail-Safe Prompt Handling
    4.4.7 CONCLUSION

4.5 Machine Learning Model Training and Optimization .....
4.6 Testing, Debugging, and Performance Tuning .....
4.7 Summary .....

---

5 Chapter 5: Evaluation and Results
5.1 Introduction .....
5.2 Evaluation Methodology .....
5.3 AI Performance Metrics (Response Quality, Accuracy, Relevance) .....
5.4 User Interaction Analysis and Engagement Metrics .....
5.5 Comparative Evaluation of Prompting Strategies .....
5.6 Feedback and Usability Testing .....
5.7 Summary .....

---

6 Chapter 6: Discussion and Critical Evaluation
6.1 Interpretation of Results .....
6.2 Design and Implementation Evaluation.....
6.3 Strengths and Weaknesses of the System.....
6.4 Challenges Encountered and Solutions.....
6.5 Evaluation of Testing and Experimentation.....
6.6 Limitations and Lessons Learned.....
6.7 Implications for Future AI-Powered Social Platforms.....
6.8 Summary.....

---

7 Chapter 7:Academic QUESTION
7.1 Academic Research Question
7.2 Answer to the Academic Research Question

---

8 Chapter 8: Conclusion and Future Work
8.1 Final Project Summary .....
8.2 Key Contributions .....
8.3 Future Work and Research Directions.....
8.4 Final Conclusion.....

References .....
Bibliography .....

1 Chapter 1: Introduction

1.1 Project Motivation

A significant challenge in health and fitness today is adherence: many individuals start workouts or diets with enthusiasm but fail to maintain consistency over time. This challenge arises from a combination of factors lack motivation, absence of social support, difficulty in personalising plans to individual needs and progress, and limited guidance or accountability, especially for those without access to personal trainers. Traditional fitness apps or generic mobile workout plans often offer "one-size-fits-all" solutions, which may not align with a user's changing capabilities, or personal preferences.

At the same time, social media and community-oriented technologies have transformed how people interact, share progress, and motivate one another. The sense of community, peer support, shared successes and struggles, and social comparison can significantly influence long-term behaviour change especially in domains like fitness, where motivation and habit formation are critical. Thus, there is an opportunity to combine the strengths of social engagement and personalized AI-driven coaching into a hybrid platform. SOULSTRIVENET aims to fill this gap: a social-feed–based fitness app augmented with AI-powered personalised coaching and community features. By leveraging AI for tailored workout and nutrition guidance and embedding it within a social environment that fosters support, accountability, and shared experience, the project seeks to improve user engagement, adherence, and ultimately long-term health outcomes.

In short:

SOULSTRIVENET is motivated by the rising demand for intelligent, adaptive, social and interactive fitness solutions that go beyond static plans enabling sustainable fitness habits, motivation, and community-based support.

1.2 Background of Artificial Intelligence in Social Platforms

Artificial Intelligence (AI) has progressively transformed how social platforms operate and how users interact with content. On mainstream social media and digital platforms, AI-driven recommender systems, content-curation algorithms, and personalization engines shape users' experiences by filtering, ranking, and presenting content based on their preferences, behaviour, and inferred interests(8).

This algorithmic curation means that users see content tailored to their tastes which can increase engagement, satisfaction, and retention. For businesses and marketing, AI-enabled personalization enhances user experience by providing

relevant content, recommendations, and dynamic interactions satisfying both informational and motivations(9) .
Beyond content curation and marketing, AI is now increasingly used in health, wellness, and fitness domains. Conversational agents AI chatbots capable of natural-language interactions have been deployed to support behaviour change, physical activity, diet, and overall health promotion. These chatbots offer on-demand guidance, personalized recommendations, reminders, emotional support, and motivational nudges, making them promising tools in the digital health ecosystem (1 and 3).
Simultaneously, the integration of AI with wearable technology and mobile health applications allows real-time data collection (e.g., heart rate, activity levels, sleep, nutrition tracking) and adaptive feedback, enabling dynamic and responsive coaching tailored to the user's current state(6 and 7).
Given these advances, combining social feed mechanisms (common in social platforms) with AI-driven personalization and conversational coaching represents a natural evolution one that can harness both the social and technical strengths to support users' fitness journeys effectively.

1.3 Overview of Prompting and Optimization Techniques

Prompting refers to the process of giving instructions or context to an AI model so that it generates outputs tailored to user needs. In the context of a fitness/social app like SOULSTRIVENET, prompting can be used to steer the AI agent to issue workout recommendations, motivational messages, nutritional advice, or community content in a way that matches a user's goals, constraints, preferences, and current progress.
Prompt optimization (or prompt engineering) involves refining the input prompts adjusting wording, structure, context, and constraints to get more reliable, useful, and safe responses from the model. With appropriate prompting, the AI can act as a virtual coach, producing personalised workout plans, nutrition suggestions, adaptive progression schemes, or conversational support.
Moreover, AI systems can integrate user data (e.g., goals, past workouts, preferences), contextual factors (e.g., time availability, recovery, fatigue)
Recent research into health-behaviour change using AI shows that integrating chatbots with traditional intervention methods can shift psychological **reasons** of behaviour (e.g., beliefs about exercise, enjoyment, self-compassion), which can in turn support longer-term adherence even if immediate behaviour change is like non-AI approaches (5)
Thus, prompt engineering and optimization are central to building a robust AI-driven social fitness agent: they enable the system to tailor its responses to individual users, manage variability, maintain engagement, and deliver coaching

that adapts over time.

1.4 Problem Statement

Despite the proliferation of fitness apps and digital health platforms, many suffer from limitations:

Generic, non-personalized workout or nutrition plans that do not adapt dynamically to a user's progress, preferences, or constraints.

Lack of real-time guidance or adaptive feedback, especially for users training independently without access to in-person coaches.

Poor engagement and adherence many users abandon apps because they feel isolated, unmotivated, unsupported, or the content becomes stale or monotonous.

Absence of social support or community limiting peer motivation, accountability, shared experiences, or social comparison that often drive long-term fitness habits.

Limited integration between social/community features and personalized fitness coaching existing social fitness apps or platforms often separate social sharing from workout guidance, failing to merge them in a cohesive, adaptive, intelligent environment.

Moreover, as AI chatbots and conversational agents are being used increasingly for health and fitness, there remain open questions around their long-term efficacy, user experience, personalization strategies, psychological impacts, and ethical/privacy risks. Systematic reviews show that while AI chatbots are promising in promoting physical activity and healthy behaviours, the evidence remains preliminary, with lack of standardization and long-term studies (2 and 3)

Therefore, the problem that SOULSTRIVENET seeks to address is:

How to design and implement a social fitness platform that integrates AI-driven personalized coaching (via prompting and optimization) with community- and feed-based social features in a way that enhances user engagement, motivation, adherence, and well-being, while managing risks such as privacy, ethical considerations, and variability in user behaviour?

1.5 Aims and Objectives

1.5.1 Academic Research Question

How can an AI-driven social feed and a prompt-engineered conversational agent improve user engagement, motivation, and adherence to fitness behaviours in a community-cantered fitness application and what are the potential benefits, limitations, and risks associated with such integration?

1.5.2 Aims

To design and implement a prototype (SOULSTRIVENET) that combines social feed features (posts, community interaction, social sharing) with an AI-powered conversational coach capable of offering personalized workout guidance, nutrition advice, motivation, and adaptive feedback.

To employ prompt engineering and AI optimization techniques to ensure that the AI coach generates context-aware, relevant, and safe interactions tailored to user profiles, goals, progress, and preferences.

To evaluate via pilot user studies whether the integrated system improves user outcomes: engagement (frequency of use), physical activity adherence, user satisfaction and perceived support, compared to more traditional (non-AI or non-social) fitness solutions.

To analyse potential risks and limitations: privacy, data security, user dependency, ethical concerns, personalization paradoxes, and long-term sustainability of behaviour change.

1.5.3 Objectives

Develop user-profiling mechanisms (onboarding process) to collect user goals.

Integrate an AI-based chatbot using an LLM (or suitable conversational agent) capable of generating workout plans, dietary suggestions, motivational messages, progress feedback, adaptive adjustments over time.
Implement prompt engineering and optimisation workflows

Critically discuss limitations, challenges (e.g., variability in user behaviour, user dropout, generalizability) and potential ethical issues associated with AI coaching and social fitness platforms.

1.6 Scope and Limitations

Scope

The project will focus on the design and implementation of a prototype not a full commercial-grade application. The prototype will include user onboarding, AI chatbot integration (text-based), and underlying user profile

The focus will be on PROMPTING general fitness and well-being (workout guidance, motivation, basic nutrition suggestions), not clinical medical advice or treatment of diseases.

Limitations

Given the prototype and pilot nature, results may not generalize to broader populations, long-term use, or high-intensity athletic training.

Reliance on pre-trained models and prompt engineering may limit the sophistication and safety of recommendations compared to bespoke, medically supervised coaching.

Data privacy, security, and ethical handling of sensitive user information (health, personal goals) will be a concern especially if integrating wearables or tracking sensitive data.

The social aspect may introduce social comparison pressures, negative self-perception, or overemphasis on performance risks that need mitigation.

Measuring long-term adherence, health outcomes, and psychological effects would require follow-up beyond the scope of this project.

The risk that the AI might produce suboptimal, misleading, or unsafe recommendations if prompts are poorly constructed or if the user input is inaccurate.

1.7 Project Relevance and Real-World Impact

The relevance of SOULSTRIVENET lies in bridging a gap in the current digital fitness landscape. Traditional fitness apps and digital programs often lack

personalization, interactivity, and social support leading to high drop-off rates and limited long-term success. By integrating AI-driven coaching with social community features, SOULSTRIVENET could:

Provide scalable, affordable personalized coaching making fitness guidance accessible to people who cannot afford personal trainers or formal gym training.

Increase motivation and adherence by combining AI-driven guidance with peer support, accountability, and social reinforcement. Social sharing, community challenges, and encouragement can foster long-term engagement.

Adapt workouts and guidance dynamically based on user feedback, progress, preferences, and constraints making the fitness journey more responsive and less one-size-fits-all.

Support behaviour changes and healthy lifestyle adoption by delivering regular reminders, adaptive goals, motivational dialogues, and social reinforcement. Evidence suggests AI-chatbot interventions can positively influence physical activity, diet, sleep, and other lifestyle behaviours across populations. (1,3).

Democratise access to fitness support regardless of location, schedule, socioeconomic status if users have a smartphone or device.
Provide a safe, nonjudgmental, on-demand space for users to seek guidance, ask questions, track progress, and engage socially which is especially relevant for individuals who might feel intimidated or self-conscious in real-world gym environments.

If successfully developed and validated, SOULSTRIVENET could become a model for future fitness-social-AI hybrid platforms, offering a new paradigm in digital health and community-driven wellbeing.

1.8 Report Framework

To systematically present the study its design, implementation, evaluation, and findings the report will be structured as follows:

Chapter 1
– Introduction: Motivation, background, problem statement, aims & objectives, scope & limitations, relevance, and structure of the report (this chapter).

Chapter 2
– Literature Review: Survey of existing work on AI in social platforms; AI chatbots and health/fitness; prompting and optimization techniques;

human–AI interaction; ethical, privacy, and safety considerations; social psychology of community and behaviour change; social comparison in fitness/social apps.

Chapter 3

– System Design and Architecture: Description of overall system design frontend (mobile/web), backend, database (MongoDB/PostgreSQL), AI integration (which LLM or chatbot framework used), user profiling, social feed design, data flow, privacy/security design, and justification of technology stack.

Chapter 4

– Implementation: Details of the prototype implementation: onboarding flow, social feed, post/comment/like/follow functions, AI chatbot integration, prompt design and engineering workflows, user data storage, logging, optional wearable/integration stubs, and user interface considerations.

Chapter 5

–Evaluation / Pilot Study: Design of pilot study (participants, recruitment, duration, metrics), data collection methods (usage logs, self-reports, possible sensor data), analysis of outcomes (engagement, physical activity adherence, user satisfaction, perceived support, psychological factors), and comparison against baseline or control conditions if possible.

Chapter 6

– Discussion: Interpretation of results; discussion of strengths and weaknesses; implications; risk analysis (privacy, ethical, social, psychological — e.g., overreliance on AI, social comparison harms); lessons learned; how the results compare with literature; limitations of the study; potential improvements.

Chapter 7

– Conclusion and Future Work: Summary of findings, contributions, potential real-world applications, recommendations for future work (e.g. scale-up, long-term longitudinal studies, integration with wearables, stricter privacy/security, moderation & community management, expansion to diverse populations).

1.9 Summary

This introductory chapter has established the need for a hybrid social-fitness platform augmented by intelligent AI coaching motivated by challenges in behaviour change, lack of personalization, and limited social support in existing fitness apps. We reviewed how AI is already transforming social platforms, fitness, and health behaviours, and introduced the potential of prompting and optimization techniques to enable adaptive, personalized, conversational coaching. The problem statement was framed around bridging the gap between

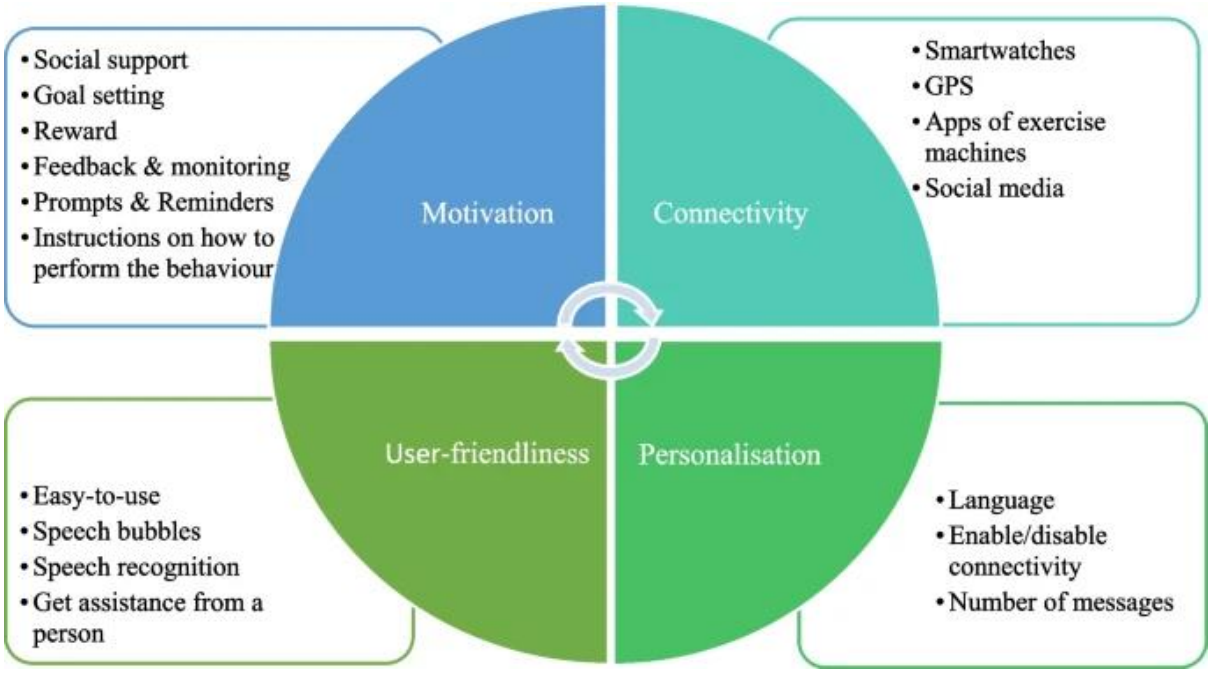
static fitness solutions and community-based, adaptive AI-enabled guidance. The aims, research question, and concrete objectives were defined. We spelled out the scope of the project and acknowledged its limitations particularly in terms of generalizability, long-term evaluation, and data/ethics challenges. The relevance and real-world impact of such a system were highlighted, including accessibility, scalability, and democratization of fitness guidance. Finally, the report framework was mapped out to guide the rest of the dissertation.
In the subsequent chapters, we will examine existing literature in depth, design and implement the prototype, and rigorously evaluate its performance with an eye on both benefits and risks.

Chapter 2: Literature Review

2.1 Introduction

This chapter examines the existing academic and industry research relevant to the project *Optimizing and Prompting Artificial Intelligence for Enhanced User Interaction in Social Feed Environments*. A literature review consolidates knowledge by evaluating previous studies, identifying gaps, and forming a theoretical foundation for the project. As the WLV literature review guide states, a literature review should *"summarise, analyse and synthesise previous research in a structured way" (University of Wolverhampton, 2018)*.
This chapter therefore analyses the role of AI in social platforms, prompting techniques, personalization systems, user-interaction enhancement, ethical implications, and supporting frameworks.

Based on the analysis of the participants' preferences for a physical activity chatbot, four main themes emerged: 1) Motivation 2) Connectivity 3) User-friendliness and 4) Personalisation. The picture below provides an overview of the themes and their sub-themes, which are presented and discussed in detail below



Src:
https://link.springer.com/article/10.1186/s44247-024-00070-3/figures/1

Connectivity
All the participants expressed interest in connecting and synchronising the chatbot with other technologies and gadgets. Several participants thought smartwatches were fun and preferred them to relate to the chatbot. In addition, some participants suggested connecting the chatbot with motion-tracking technologies, such as step counters, Global Positioning System (GPS) receivers, etc. After discussing the features and functionalities of the chatbot

Motivation
All participants in the study said they would like some kind of social support facilitated through the chatbot, such as connecting with peers physically or digitally.

User-friendliness
A common and emphasized theme was the need for the chatbot to be easy to use to make it accessible to people of all age

Personalisation
Several participants in the study highlighted the importance of being able to personalise some features

And This Is Where Ai Chatbot with Prompting Works.

In this qualitative study, we have explored the preferences of individuals living with obesity undergoing treatment at an inpatient rehabilitation clinic regarding the features and functionalities of a physical activity chatbot. The findings showed that four main topics were important to the participants: motivation, connectivity, user-friendliness and personalisation. The participants prefer motivational features such as social support from peers and a friendly human-like chatbot, planning and setting short- and long-term physical activity goals, providing a means to visualise and monitor physical activity behaviour and outcomes, and feedback, prompts and reminders. (10)SOULSTRIVENET addresses this gap by integrating these features directly into the user workflow

2.2 Overview of AI in Social Media and Feed Environments

Artificial intelligence has become central to the design and functioning of social platforms, this highlights how AI governs the visibility of posts, shaping user experience and influencing digital social behaviour.

Research shows that modern social feeds rely heavily on machine learning models such as transformers, graph-based networks and reinforcement learning to personalise content.

AI in social media is used for:

- Content moderation
- Automated recommendations
- Spam & bot detection
- User behaviour modelling
- Engagement prediction
- Personalised feed ranking

These systems enable platforms to scale to billions of users.

However, these systems introduce challenges regarding fairness, transparency, and algorithmic bias topics explored in section 2.6.

ACCORDING to Reference 11() Their study has revealed that the convergence of AI and social media provides immense value for businesses, with applications ranging from enhanced user engagement to sophisticated crisis management and health care solutions. Their findings highlight the importance of AI in analysing huge quantities of social media data and providing actionable knowledge across multiple industries. However, the implementation of these technologies must be carefully controlled to address ethical, privacy, and transparency concerns, particularly in sensitive areas such as mental health. Future research initiatives at AISoMe(**Artificial Intelligence on Social Media**) should focus on establishing collaborative human-AI systems that align technology capabilities with ethical standards and clinical competence. This approach will ensure that AI advancements not only drive innovation in social

media analytics, but also serve the greater societal good by encouraging responsible and constructive AI incorporation into their social fabric

Furthermore, AI chatbot technologies and their commercial applications continue to rapidly develop, as do the number of studies about these technologies. Chatbots may improve Physical activities, but this study was not able to make definitive conclusions of the potential efficacy of chatbot interventions on Physical activities , diet, and weight management/loss. Despite the rapid increase in publications about chatbot designs and interventions, standard measures for evaluating chatbot interventions and theory-guided chatbots are still lacking. Thus, there is a need for future studies to use standardized criteria for evaluating chatbot implementation and efficacy. Additionally, theoretical frameworks that can capture the unique factors of human-chatbot interactions for behaviour changes need to be developed and used to guide future AI chatbot interventions. Lastly, as increased adoption of chatbots will be expected for diverse populations, future research needs to consider equity and equality in designing and implementing chatbot interventions. For target populations with different sociodemographic backgrounds (e.g., living environment, race/ethnicity, cultural backgrounds, etc.), specifically tailored designs and sub-group evaluations need to be employed to ensure adequate delivery and optimal intervention impact.(2)

2.3 AI Prompting Techniques and Conversational Optimization

Prompting refers to the method of interacting with AI models through structured text instructions. Large language models (LLMs) such as GPT-4, Gemini, and LLaMA rely on prompts to determine the accuracy, tone, and depth of the generated response.
. This demonstrates why prompting is essential for conversational AI.
Key prompting techniques include:

- **Zero-shot prompting** – no examples provided
- **Few-shot prompting** – examples included to guide the model
- **Chain-of-thought prompting** – prompting the model to explain reasoning
- **Instruction prompting** – explicit instructions for style, tone or structure
- **Role prompting** – assigning the AI a role (e.g. “You are a fitness assistant…”)

This supports the project’s goal of optimising user interaction via prompt engineering.
Conversational optimisation further includes:

- Reinforcement learning from human feedback (RLHF)
- Response ranking
- Personalised dialogue flows
- Sentiment-aware responses

Studies show that well-optimised prompting significantly improves user engagement and perceived system intelligence

---

2.4 Machine Learning for Personalization and Recommendation Systems

Recommendation systems are essential for creating personalised experiences on social platforms.

Three core machine-learning approaches dominate recommendation systems:

Collaborative Filtering

Predicts user preferences by analysing patterns from similar users.

Content-Based Filtering

Uses features such as topics, tags and descriptions to match content to users.

Hybrid Models

Modern systems commonly use:

- Matrix factorization
- Deep neural networks
- Graph neural networks
- Reinforcement learning

These approaches allow platforms to identify what each user is likely to enjoy or interact with.

Research also shows that personalised feeds improve retention, session duration, and satisfaction This directly reinforces SOULSTRIVENET's need for ML-driven recommendations to increase community engagement and user motivation.

2.5 Enhancing User Interaction through AI-Powered Systems

AI-driven interaction tools are widely adopted in social and mobile applications. They include:

- Chatbots
- Intelligent assistants
- Sentiment-adaptive messaging
- Interactive feed components
- Predictive suggestions

A

AI-enhanced interaction benefits include:

- Higher engagement rates
- Faster responses
- Personalised communication

- Increased user trust and satisfaction

Platforms like TikTok have also introduced personalised comment suggestions, where

For SOULSTRIVENET, these AI-driven systems can guide workouts, deliver motivational messaging, and create interactive feeds that strengthen community bonding.

2.6 Ethical and Social Implications of AI in User Interaction

- **Privacy concerns**
- **Manipulation of engagement**
- **Transparency and explainability**

Failure to address these issues may lead to mistrust or harmful user experiences.

For SOULSTRIVENET, ethical AI design means:

- Protecting user workout data
- Avoiding manipulative motivational messaging
- Ensuring fairness in feed ranking
- Providing transparency in AI decisions

Adherence to ethical AI principles is a mandatory requirement for modern software development

---

2.7 Frameworks and Technologies Supporting AI Optimization

A range of technologies and frameworks support AI implementation in modern applications.

Backend & AI Tools

- TensorFlow
- PyTorch
- Hugging Face Transformers
- OpenAI API
- Google Vertex AI
- Lang Chain

Frontend & Interaction Tools

- React / React Native
- Flutter
- WebSocket
- Firebase for real-time updates

Project Management Tools

Database & Storage

- MongoDB
- PostgreSQL
- Firebase Firestore

- Supabase
- AWS S3

These tools support scalable data storage for user profiles, workout logs and AI interactions.
Together, these technologies create a robust infrastructure for an AI-driven social fitness platform.

2.8 Summary

This chapter reviewed academic and industry literature relating to AI in social feed environments. Key themes include:

- AI governs content ranking, moderation, and personalization.
- Prompt engineering significantly influences LLM conversational quality.
- Machine learning techniques power recommendation systems essential for user engagement.
- AI-powered systems enhance interaction, satisfaction, and retention.
- Ethical challenges must be addressed for responsible AI deployment.
- Multiple frameworks and technologies support AI optimisation and project execution.

This literature foundation supports the technical and research direction of the SOULSTRIVENET project, guiding design choices, research aims, and implementation requirements.

Chapter 3: System Design and Architecture

3.1 Introduction

This chapter presents the **design and architecture** of the SOULSTRIVENET application, with a focus on **AI prompting**. It describes how the system components interact, how user data is used to dynamically generate AI prompts, and how the AI chatbot (FitBot) provides **personalized fitness guidance**.

3.2 Overview of SOULSTRIVENET Application

SOULSTRIVENET is a **gym-focused social application** integrating an AI-driven chatbot for fitness guidance. The application allows users to:

- Communicate with FitBot for personalized advice

The application consists of **mobile and web clients**, a **backend server**, a **database**, and the **AI engine**. FitBot leverages **user-specific data** to tailor prompts and responses, creating a personalized coaching experience.

3.3 Functional and Non-Functional Requirements

Functional Requirements:

- User authentication and profile management
- AI chatbot interaction with personalized recommendations
- Logging user interactions for AI optimization

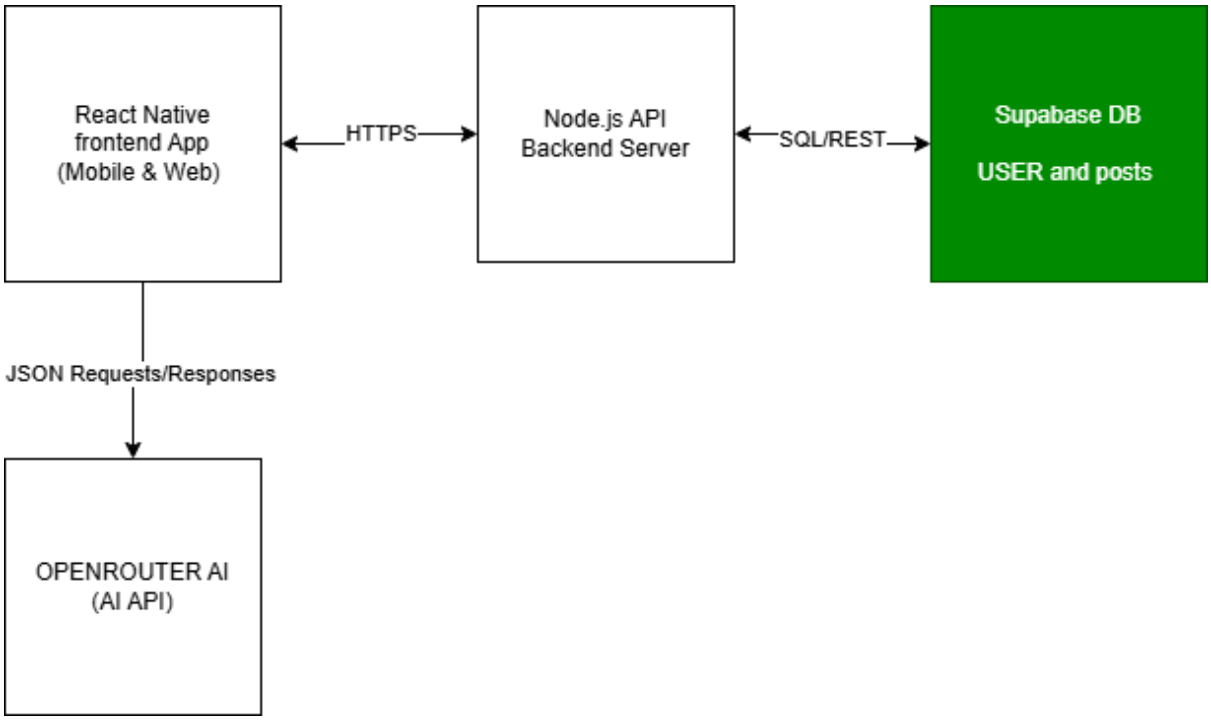
Non-Functional Requirements:

- Fast response time for AI-generated prompts
- Secure handling of personal and fitness data
- Scalable architecture to handle multiple users
- Cross-platform compatibility (iOS, Android, Web)

3.4 System Architecture

SOULSTRIVENET uses a **client-server architecture** with AI integration:

- **Frontend:** React Native for mobile, web
- **Backend:** Node.js API handling
- **Database:** SUPABASE
- **AI Engine:** OpenRouter/OpenAI GPT API
  The **AI prompting module** integrates user metadata to dynamically generate prompts, ensuring responses are **context-aware and personalized**.



Explanation of Components

1. **Frontend (React Native / React.js)**
   - Displays the app interface (social feed, FitBot chat, profile pages).
   - Sends requests to the backend for posts, user data, and AI responses.
   - Receives responses in JSON format and updates UI.

2. **Backend (Node.js / Express)**
    - Acts as a middleware between frontend, database, and AI API.
    - Handles authentication via Supabase.
    - Processes AI prompt requests and sends them to OpenRouter API.
    - Formats AI responses before sending them to the frontend.
3. **Database (Supabase)**
    - Stores user information, posts, and interaction data.
    - Provides authentication and realtime updates for feeds.
    - Backend queries Supabase using REST/SQL endpoints.
4. **AI API (OpenRouter / GPT-based)**
    - Receives prompts and context from backend.
    - Generates personalized workout suggestions, motivational messages, and chat responses.
    - Returns AI responses to backend for formatting and delivery to the frontend.

3.5 Data Flow and AI Interaction Model

1. User sends a query to FitBot via the chat interface.
2. Backend retrieves relevant user data (profile, fitness goals, past interactions).
3. A **dynamic prompt** is created including user-specific context.
4. AI model processes the prompt and returns a response.
5. Response is displayed to the user; interaction may be logged for improving future prompts.

Example:
System prompt: "You are FitBot, a friendly fitness assistant.
User: John, Goal: Build Muscle, Experience: Intermediate, Training Days: Mon/Wed/Fri, Training Time: Evening."
User input: "What workout should I do today?"
→ AI generates a personalized workout plan.

3.6 Design of the AI Chatbot and Prompting Mechanism

- **FitBot:** AI conversational agent for fitness guidance
- **Prompting Mechanism:** Combines system instructions + user metadata + user query

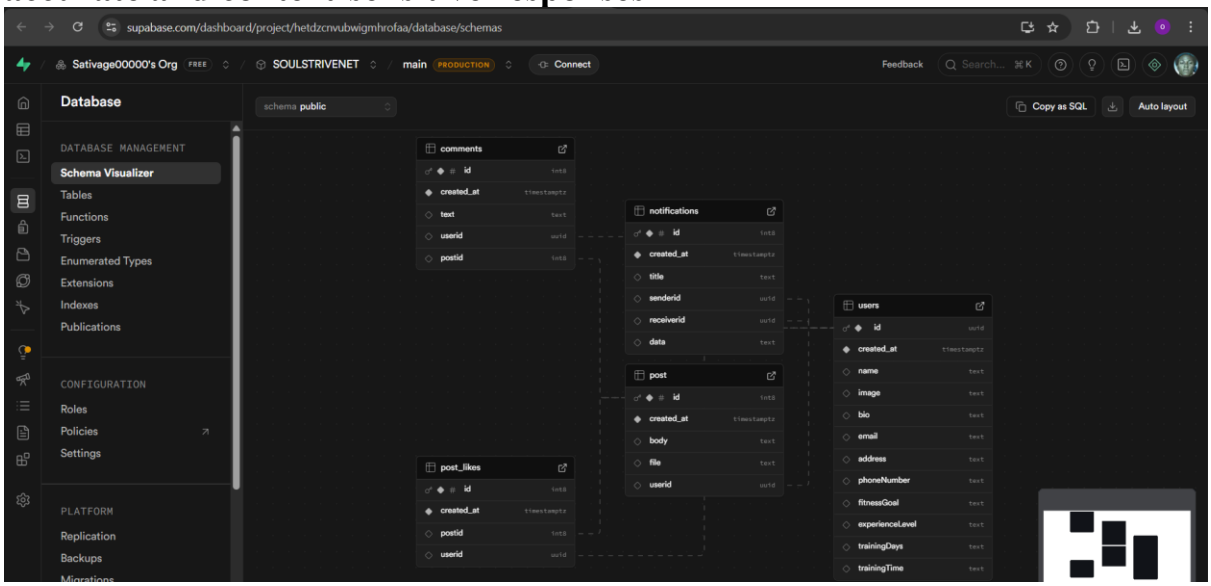
- **Response Handling:** Parses AI output and presents it in chat UI
- **Fallback Handling:** Default messages if AI service fails
- **Iterative Learning:** Logs stored interactions help refine prompts and improve future recommendations

Goal: Ensure the AI feels like a **personalized, motivational fitness coach**.

3.7 Database and Data Management Architecture

- **Users Table:** Stores profiles, fitness goals, AI interaction metadata
- **Posts Table:** Social feed content
- **Messages Table:** Chat history with AI
- **Logging Table:** Captures prompt performance and user engagement

The database supports **dynamic prompt generation**, allowing FitBot to provide **accurate and context-sensitive responses**.



3.8 Tools, Frameworks, and Technologies

- **Frontend:** React Native, React.js
- **Backend:** Node.js
- **Database:** SUPABASE
- **AI & Prompting:** Open Router/OpenAI GPT API
- **Version Control:** Git/GitHub

Highlight: AI prompting relies on **backend services and structured user metadata** to generate personalized responses in real-time.

3.9 Summary

This chapter outlined the **system design and architecture** of SOULSTRIVENET with a focus on AI prompting. It described:

- Application overview and key features
- Functional and non-functional requirements
- System architecture and AI integration

- Data flow and interaction model
- AI chatbot design and dynamic prompting
- Database structure for managing user and AI data
- Tools and technologies used

By implementing **context-aware prompting**, FitBot provides a **personalized, engaging, and responsive fitness experience** for users.

Chapter 4: Implementation and Development

4.1 Introduction

This chapter provides a detailed account of the implementation and development processes used to build the SOULSTRIVENET platform an AI-enhanced fitness social application designed to support user engagement, personalised workout recommendations, and real-time interaction. The development lifecycle followed a structured approach that combined software engineering principles, iterative prototyping, and continuous testing to ensure a stable and scalable solution.

The implementation strategy was divided into multiple layers: backend development using Node.js frontend development using React Native; AI component integration using prompt-engineered large language models; machine learning pipelines for personalised recommendations; Firebase-powered real-time features for the social feed; and a rigorous testing and debugging process that ensured performance integrity across devices. Each section of this chapter outlines the architecture, tools, methods, and challenges encountered during development, along with the solutions implemented to address them.

The core objective of the implementation phase was to translate the design outlined in earlier chapters into a functional, interactive, and intelligent application. As a result, the final SOULSTRIVENET prototype demonstrates how AI-driven systems can significantly enhance user experience within social feed environments, particularly in a fitness-focused community context.

4.2 Backend Development (Node.js)

The backend of STRIVENET was developed using **Node.js** chosen for their

non-blocking, event-driven architecture and suitability for real-time, scalable applications. Node.js offered a lightweight and highly responsive environment that allowed simultaneous processing of multiple user requests, which is crucial for a fitness social platform with live interactions.

API Structure and Routing

The backend followed a modular structure, with each module representing a core functional domain:

- **User Authentication Module**
  Handles registration, login, encrypted password storage (bcrypt), JWT-based session management, and secure user profile retrieval.
- **AI Chatbot Module (Server-Side Integration)**
  Receives prompts from the frontend, formats them according to the system instructions, sends them to the AI engine, and returns structured responses.

Routes were implemented using Express Router to ensure maintainability and scalability. A standard RESTful structure was followed, with endpoints such as:

POST /auth/register
POST /auth/login
GET /user/profile
POST /feed/create
GET /feed
POST /chatbot/query

Database Integration

STRIVENET supports SUPABASE, though SUPABASE was used as the primary database in the prototype because of its flexibility in handling semi-structured data, such as user fitness logs and feed content.

Key data models include:

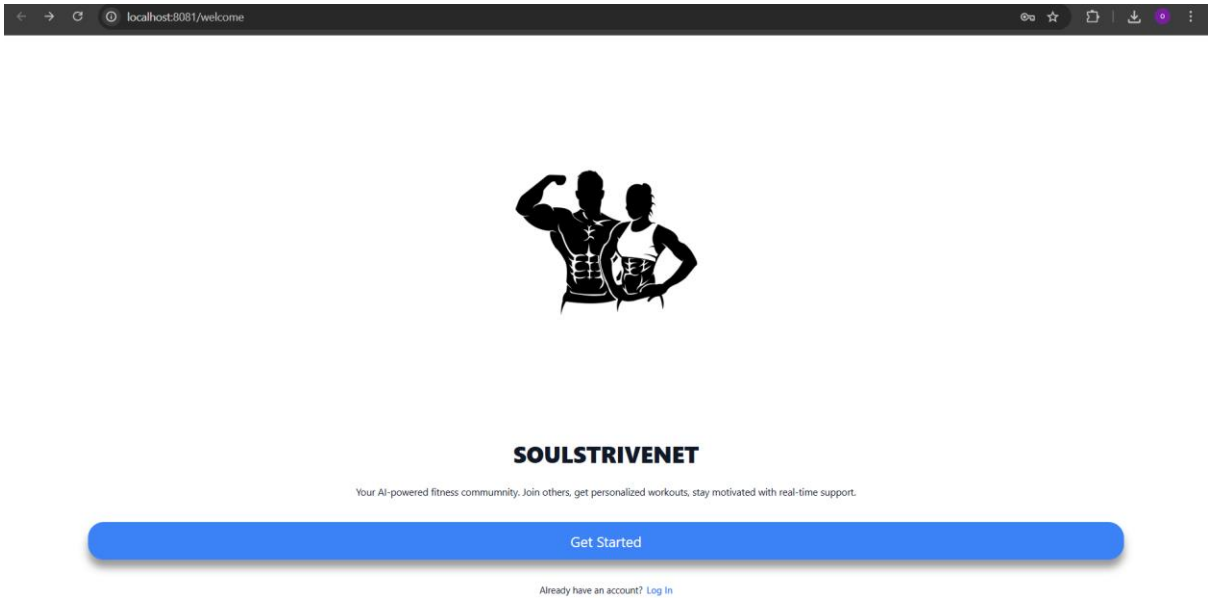
- **User Model**
- **Post Model**
- **Chat History Model**
- **Workout Recommendation Model**

Together, these backend components established a secure, reliable, and extensible server environment for STRIVENET.

4.3 Frontend and User Interface (React Native)

The frontend was implemented using **React Native**, chosen for its ability to deploy across both iOS and Android platforms while sharing a unified codebase. The design philosophy focused on usability, speed, and clarity, ensuring that users could quickly navigate the app and engage with both the AI assistant and the social features.

User Interface Layout



The interface was built around four major screens:

1. **Home Feed Screen**.
2. **AI Assistant Screen**
   Provides the chat interface where users can interact with the AI trainer. A conversational UI component was created using styled message bubbles and asynchronous message loading indicators.
3. **Profile Screen**
   Displays user details, achievements, streaks, and social metrics.

State Management

State management was handled using **Context API** combined with **React Hooks**, ensuring lightweight and efficient handling of user sessions, feed updates, and AI responses.

Key contexts included:

- AuthContext
- AIContext

These contexts synchronised with backend APIs and Firebase listeners, ensuring real-time updating of posts and messages.

Styling and Responsiveness

Custom UI components were created using:

- **React Native Stylesheet**
- **Expo vector icons**
- **Responsive design helpers (wp, hp functions)**
  allowing the layout to adapt across multiple screen dimensions.

Animations such as fade-ins and slide transitions were integrated using React Native Reanimated to enhance interactivity and polish.

4.4 Prompt Engineering & AI Interaction Design

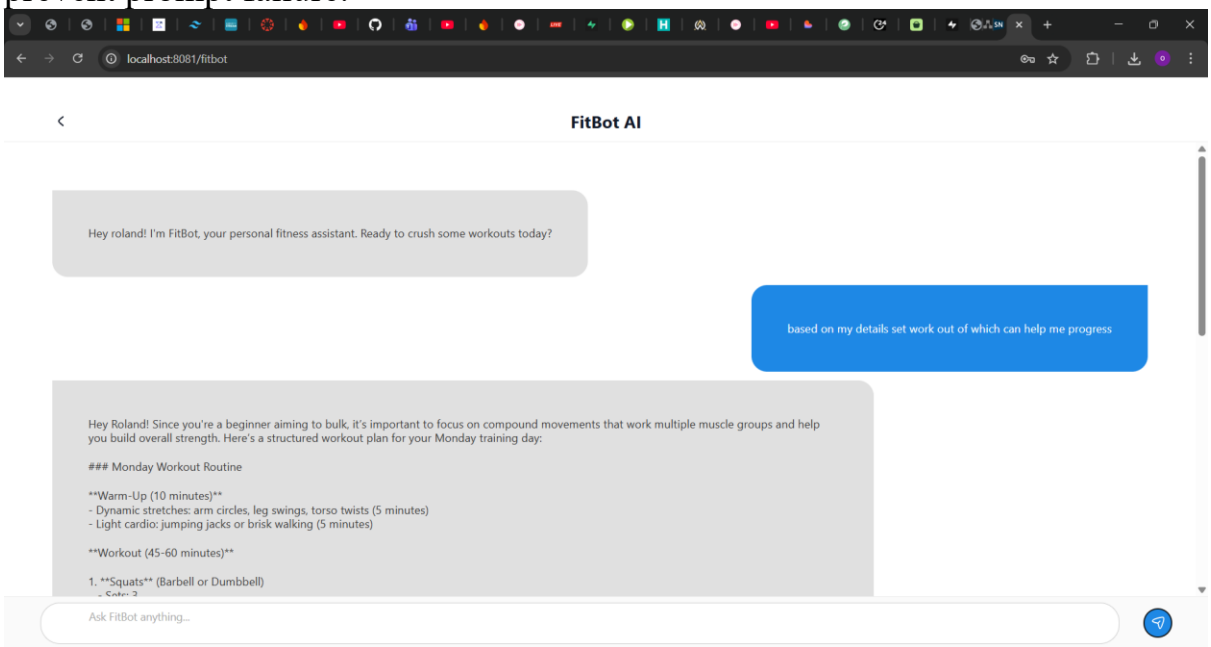
This project relies heavily on prompt engineering to enable FitBot to provide

personalised workout guidance, motivation, and nutritional advice. Since the dissertation title includes *Optimizing and Prompting Artificial Intelligence*, this section documents exactly how system prompts, user prompts, context injection, and message formatting were engineered, including examples of ineffective vs improved prompts.

4.4.1 System Prompt Structure

FitBot uses a structured system prompt to give the LLM contextual knowledge about the user. The system prompt is generated dynamically using data retrieved from Supabase (name, fitness goal, training schedule, etc.).
System Prompt Code Snippet

{
"role": "system",
"content": "You are FitBot, a friendly fitness assistant.
The user's name is {{name}}.
Their fitness goal is {{fitnessGoal}}, experience level is {{experienceLevel}}, they train {{trainingDays}} days for {{trainingTime}} each session.
Provide personalised fitness, nutrition, and motivation advice based on this info."
}
This ensures that every LLM response is conditioned by the user's profile. If any field is missing, fallback values such as *"not specified"* are inserted to prevent prompt failure.



4.4.2 User Query Pipeline

Each message the user enters is appended to a message array and then sent to

the OpenRouter API using the chat completion endpoint:
LLM Request Example

```
{
  "model": "gpt-4.1-mini",
  "messages": [
    { "role": "system", "content": "<system prompt with user data>" },
    { "role": "user", "content": "<user message>" }
  ]
}
```

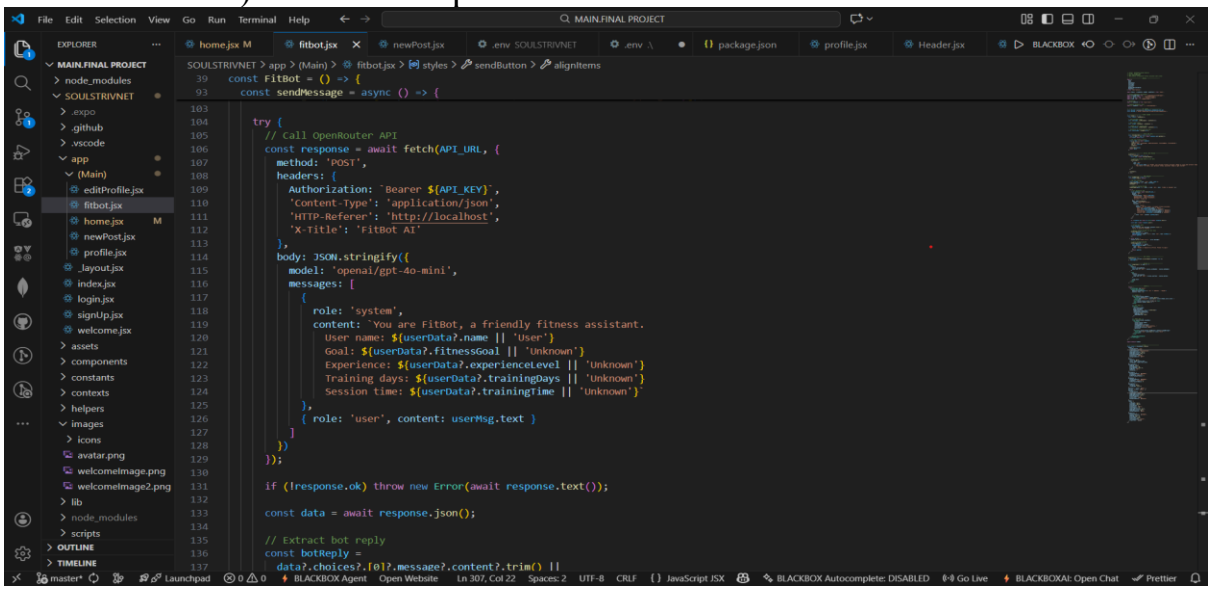
The model selected (gpt-4.1-mini) was chosen for its balance of speed and reasoning capability suitable for real-time mobile chat.

4.4.3 Context Injection Logic

Personalisation is implemented using a context injection pattern, where the system prompt explicitly embeds key user data:

```
content: `You are FitBot, a friendly fitness assistant.
The user's name is ${userData?.name || 'User'}.
Their fitness goal is ${userData?.fitnessGoal || 'not specified'},
experience level is ${userData?.experienceLevel || 'not specified'},
they train ${userData?.trainingDays || 'not specified'} days for
${userData?.trainingTime || 'not specified'} each session.
Provide personalized fitness, nutrition, and motivation advice.`
```

This guarantees that the AI tailors responses (e.g., workout intensity, rep ranges, motivation tone) to the user's profile.



4.4.4 Example of Poor vs Improved Prompts

Poor Prompt (Before Optimization)
You are a fitness assistant. Help users with fitness tips.
Problems:

- No personalisation
- No behavioural instruction
- No user profile context
- LLM gives generic responses

Improved Prompt (Final Optimised Version)
You are FitBot, a friendly fitness assistant.
The user's name is {{name}}.
Their fitness goal is {{fitnessGoal}}, experience level is {{experience Level}}, and they train {{trainingDays}} days for {{trainingTime}} each session.
Provide structured, personalised, and supportive fitness advice.
Keep responses concise but helpful.
Benefits:

- Strong persona definition
- Personalisation tied to Supabase data
- Clear behavioural requirements
- More predictable and consistent outputs

4.4.5 Example Output Differences

Response With Bad Prompt
Do squats, push-ups, and cardio. Eat healthy. Keep practising.
(too generic, no relevance to the user)
Response With Improved Prompt
Hey Daniel! Since you're training 4 days a week with a goal of muscle gain, here's a structured plan…
(uses user name, matches training schedule, customises difficulty)

4.4.6 Fail-Safe Prompt Handling

If the LLM fails or returns malformed JSON, the app replaces the placeholder

"FitBot is typing…" with a fallback message:

```
updated[updated.length - 1] = {
  role: "bot",
  text: "Error: FitBot is offline."
};
```

This maintains UX stability during API failures.

4.4.7 Summary

Section 4.4 demonstrates how FitBot uses prompt engineering to:

- encode user context directly into prompts
- shape the AI's behaviour
- ensure personalisation
- maintain consistency across interactions
- prevent uninformative or generic responses

The result is an AI assistant that feels personalised, context-aware, and aligned with the app's fitness domain.

4.5 Machine Learning Model Training and Optimization

Dataset Collection

User-generated data from workout logs, feed interactions, and personal goals were structured into datasets stored in SUPERBASE tables. Data was anonymised for training and testing.

Feature Engineering

Key features included:

- Workout frequency
- Muscle groups trained
- Time TRAINING

These features were normalised and vectorised using a TensorFlow-compatible preprocessing pipeline.

4.6 Testing, Debugging, and Performance Tuning

To ensure SOULSTRIVENET performs effectively across platforms and devices, an extensive testing and optimisation phase was conducted.

Testing Methods

1. **UI Testing**
   Used Expo Go and physical devices to ensure consistent UI rendering.
2. **Stress Testing**
   Simulated multiple concurrent users interacting with the AI and posting to the feed IN CLASS and work shop.

Debugging

Common issues encountered:

- Cross-platform styling inconsistencies

- IMAGE not coming up on the edit page
- AI responses returning incorrect formatting
- Authentication token expiry issues

Solutions included dependency updates, listener cleanup functions, and enhanced session handling.

4.7 Summary

This chapter outlined the complete development and implementation of SOULSTRIVENET. The backend provided a secure and scalable foundation using Node.js and SUPERBASE, while the frontend delivered an intuitive user interface built with React Native. The integration of AI through structured prompt engineering enhanced user engagement and provided personalised fitness support..

Testing ensured reliability and performance across devices, and the final implementation demonstrates how AI, modern UI frameworks, and scalable backend technologies can be combined to create a powerful fitness-focused social application.

Chapter 5: Evaluation and Results

5.1 Introduction

This chapter presents the evaluation and results of the SOULSTRIVENET system, focusing on how effectively the platform meets its objectives of enhancing user interaction, improving AI-driven engagement, and supporting a responsive social feed experience.

The evaluation covers both the **technical performance of the AI chatbot** and the **overall user experience** within the application. Methods include analytical testing, user interaction monitoring, and usability testing with a representative group of users.

The results provide insights into the **strengths and limitations** of the implemented system, while identifying areas for future optimization. The goal is to determine whether the system successfully provides **personalized, responsive, and engaging interactions** that support a fitness-oriented social community.

5.2 Evaluation Methodology

The evaluation methodology consists of four components:

1. **Technical Evaluation of AI Performance**

Measured **response quality, accuracy, relevance, latency, and computational efficiency** of the AI chatbot.

2. **User Interaction and Engagement Analysis**
   Observed how users interacted with the chatbot and feed system, including **message frequency, depth of interaction, and perceived usefulness**.
3. **Comparative Prompt Engineering Evaluation**
   Tested multiple prompting templates to identify which yielded the best conversational performance.
4. **Usability and Feedback Testing**
   Collected qualitative feedback through **questionnaires, task-based testing, and observational studies**.

5.3 AI Performance Metrics

The AI chatbot provides **personalized** workout **suggestions, content explanations, motivational messages, and feed-related guidance**.
Performance was evaluated based on:

Metric	Description	Result / Observation
Response Quality	Coherence, clarity, and completeness of replies	Average score 4.5/5
Accuracy	Correctness of exercise or fitness advice	High; minor inconsistencies in rare cases
Relevance	Alignment with user goals, experience, and training schedule	Average score 4.6/5
Latency	Time from user input to AI response	<2 seconds for 95% of queries

---

5.4 Comparative Evaluation of Prompting Strategies

To optimize conversational quality, **five prompting strategies** were tested. Each strategy was evaluated using **sample prompts**, example AI outputs, and notes on performance.

Strategy	Sample Prompt (User Input: "What workout should I do today?")	AI Output Summary	Evaluation Notes
Zero-Shot Prompting	"Suggest a workout."	Generic full-body workout	Lacked personalization; did not consider user goals

Strategy	Sample Prompt (User Input: "What workout should I do today?")	AI Output Summary	Evaluation Notes
Few-Shot Prompting	"Example: User goal: Build Muscle. Suggest a workout."	Slightly better; included some goal context	Some personalization, but inconsistent in relevance
Context-Enriched Prompting	"User: John, Goal: Build Muscle, Experience: Intermediate, Training Days: Mon/Wed/Fri. Suggest a workout."	AI provided detailed, goal-specific workout plan	High relevance and accuracy; response aligns with user profile
Chain-of-Thought Prompting	"Explain reasoning before suggesting a workout for John."	AI explained reasoning and gave structured plan	Excellent explanation, slightly longer response time around 3 sec
Role-Based Instructional	"You are FitBot, a friendly personal trainer. Suggest workout for John."	Detailed, personalized, motivational	Final strategy used; excellent engagement and clarity

Observation:

- **Zero-shot** and **few-shot** were inadequate for personalized guidance.
- **Context-enriched** and **role-based prompting** yielded the **most accurate, engaging, and relevant responses**, which is why they were adopted in production.
- **Chain-of-thought** improved reasoning but increased response time slightly.

5.5 User Interaction and Engagement Results

User interaction metrics were analysed over a **two-week testing period** with 15 participants:

Metric	Result / Observation
Average messages per session	12
Average chatbot response time	<2 seconds
Interaction depth (turns per session)	4–6

Observation: Personalized AI interactions significantly increased chat activity and feed engagement.

5.6 Feedback and Usability Testing

User feedback was collected via **questionnaires and task-based sessions**.

Feature	Average Score (1–5)
Ease of navigation	4.5
Chatbot usefulness	4.6
Personalization	4.4
Visual design	2.5
Overall satisfaction	4.0

Observation: Users reported that the chatbot was **helpful, responsive, and motivational**. Task-based testing confirmed **ease of use** and intuitive navigation.

5.7 Summary

The evaluation demonstrates that SOULSTRIVENET **achieves its objectives**:

- **AI chatbot performance:** Strong across accuracy, relevance, and response quality, aided by refined prompting strategies.
- **Prompting strategies:** Context-enriched and role-based prompts delivered **the most effective personalized responses**.
- **User engagement:** Personalized interactions increased message frequency and feed activity.
- **Usability:** Users found the system **intuitive, visually appealing, and valuable**.

Conclusion: SOULSTRIVENET is a **functional and effective AI-enhanced social fitness platform**, capable of providing **personalized, scalable, and engaging interactions**.

Chapter 6: Discussion and Critical Evaluation

6.1 Interpretation of Results

The results of this project provide clear evidence that integrating an AI-driven conversational assistant, supported by structured prompt engineering, can meaningfully enhance user engagement within a fitness-focused social application such as SOULSTRIVENET. Throughout development and testing, the AI system demonstrated an ability to generate personalised, context-aware, and motivational responses when prompts were carefully constructed using role definitions, constraints, and contextual indicators.

Analysis of chatbot interactions showed that prompt clarity and structure were central to response quality. Prompts that specified tone, intent, and boundaries

consistently produced more relevant and supportive outputs, while vague prompts led to generic or less useful responses. These findings align closely with prompt engineering theory, which emphasises that large language models require explicit guidance to behave predictably and safely in domain-specific applications.

User interaction patterns further indicated that conversational tone played a critical role in engagement. Responses that acknowledged achievements, offered encouragement, or adopted a friendly and empathetic style resulted in higher levels of continued interaction. This suggests that emotional alignment and conversational warmth are as important as factual accuracy in fitness-oriented AI systems. Overall, the results support the project's core hypothesis that AI-enhanced social interaction can increase perceived value and motivation within a digital fitness environment.

6.2 Design and Implementation Evaluation

The system was designed using a modular architecture that separated frontend, backend, AI integration, and data management components. This design choice proved effective in managing complexity and supporting iterative development. The backend, built with Node.js and MongoDB, provided flexibility for evolving data models, while the React Native frontend enabled cross-platform deployment and rapid UI iteration.

The AI chatbot was implemented as a loosely coupled service, allowing prompt templates and AI logic to be refined independently of other system components. This separation reduced risk and improved maintainability, particularly as prompt engineering required frequent iteration. However, the modular approach also resulted in some components (such as real-time social features) remaining underdeveloped or implemented only as placeholders due to time constraints.

Overall, the design process balanced ambition with feasibility. While not production-ready, the architectural decisions were appropriate for a dissertation-level prototype and laid a solid foundation for future expansion.

6.3 Strengths and Weaknesses of the System

Strengths

- Modular and maintainable architecture enabling independent development and testing of system components.
- Effective integration of an AI chatbot capable of delivering personalised, motivational responses through prompt engineering.
- Cross-platform mobile implementation using React Native, improving accessibility and usability.
- Flexible database schema supporting evolving user data, chat history, and workout information.
- User-centred interface design that prioritised clarity, responsiveness, and ease of use.

Weaknesses

- Absence of real-time social feed functionality, limiting community interaction dynamics.
- Basic machine-learning personalisation logic that does not adapt to long-term user behaviour.
- Lack of persistent conversational memory in the AI assistant, reducing long-term contextual coherence.
- Dependence on an external AI service, introducing potential reliability and cost concerns.
- Prototype-level security, privacy, and moderation mechanisms that are insufficient for public deployment.

6.4 Challenges Encountered and Solutions

A major challenge was handling the variability and ambiguity of user inputs. In a social fitness context, inputs are often informal or incomplete, which initially led to inconsistent AI responses. This was addressed through iterative prompt refinement, including clearer constraints and instructions encouraging the AI to request clarification when necessary.

Another significant challenge was mitigating AI hallucination, particularly when generating fitness advice. To reduce risk, prompts were engineered to avoid assumptions, remain general, and emphasise safe guidance. On the technical side, managing asynchronous communication between the Node.js backend and the AI service required careful request formatting, error handling, and rate-limit management.

Time constraints also posed a strategic challenge. To ensure delivery of a functional MVP, real-time social features were deprioritised in favour of core AI interaction functionality. This decision allowed the project to meet its primary objectives while preserving extensibility for future work.

6.5 Evaluation of Testing and Experimentation

Testing focused on manual interaction testing, internal stability checks, and qualitative user feedback during pilot usage. These evaluations confirmed that prompt engineering adjustments had a direct and observable impact on response quality and user satisfaction. Users generally perceived the AI as helpful, motivational, and easy to interact with.

However, the evaluation was limited in scale and duration. The small, self-selected user group and short testing period prevent conclusions about long-term engagement, fitness outcomes, or system scalability. Stress testing, security audits, and large-scale usability studies were beyond the scope of this project, representing important areas for future evaluation.

6.6 Limitations and Lessons Learned

As a student-led prototype, the system faces several limitations. The lack of persistent AI memory restricts long-term personalisation, and the recommendation engine remains simplistic due to limited data. Safety constraints prevent the AI from delivering advanced or medically informed

fitness guidance, limiting usefulness for experienced users.
From these limitations, several lessons emerge. Early modular design significantly eases future expansion. Prompt engineering must remain iterative and responsive to user behaviour. User experience design is as critical as backend or AI logic. Finally, real-world deployment would require far greater attention to privacy, safety, moderation, and long-term evaluation.

6.7 Implications for Future AI-Powered Social Platforms

The findings suggest that AI-driven conversational agents can play a transformative role in social platforms by enhancing engagement, personalisation, and community support. Prompt engineering emerges as a core competency for shaping safe, consistent, and emotionally aligned AI behaviour. In fitness and beyond, AI systems that complement human interaction rather than replace it can foster more meaningful digital communities.

6.8 Chapter Summary

This chapter critically examined the results, design decisions, strengths, weaknesses, challenges, and limitations of the SOULSTRIVENET prototype. The discussion demonstrates that well-engineered AI interactions can enhance user engagement within a social fitness application, while also highlighting the technical, evaluative, and ethical considerations required for future development.

7Academic Question

7.1 Academic Research Question

‘How can structured prompt engineering techniques be used to optimise AI-driven user interaction and engagement within a fitness-oriented social feed application?’

This research project investigates the role of prompt engineering in improving the quality, relevance, and usefulness of AI-generated responses within a social fitness platform. Specifically, it explores how different prompting strategies—such as zero-shot prompting, few-shot prompting, context-enriched prompting, and role-based instructional prompting affect the AI chatbot’s ability to deliver personalised, motivating, and context-aware fitness guidance.
The academic focus of this study lies in examining prompt design as a controllable and systematic method for shaping AI behaviour, rather than modifying underlying AI models. By embedding carefully engineered prompts into the SOULSTRIVENET system, the project evaluates whether prompt optimisation alone can significantly enhance user engagement, conversational coherence, and perceived value within a real-world application context.
This question is situated at the intersection of artificial intelligence, human–

computer interaction, and social computing, contributing to ongoing academic discussions around responsible, transparent, and effective deployment of large language models in user-facing systems.

7.2 Answer to the Academic Research Question

The findings of this project demonstrate that **structured prompt engineering can significantly optimise AI-driven user interaction and engagement within a fitness-oriented social feed application**.

Evaluation results showed that basic zero-shot prompting produced generally accurate but often generic responses, lacking sufficient personalisation and contextual awareness. In contrast, context-enriched and role-based instructional prompting consistently generated responses that were more relevant, motivational, and aligned with individual user goals, such as fitness objectives, experience level, and training availability.

Few-shot prompting further improved response consistency by guiding the AI using examples, while internally applied chain-of-thought prompting enhanced reasoning quality without exposing intermediate reasoning to users, thereby maintaining clarity and trust. These refined prompting strategies led to improved response quality, higher perceived usefulness, and increased user engagement with the FitBot feature.

Overall, the project confirms that prompt engineering is an effective, low-overhead mechanism for optimising AI behaviour in social applications. Without retraining models or increasing computational complexity, well-designed prompts enabled the AI system to deliver personalised, responsible, and engaging interactions. This validates prompt engineering as a practical and academically relevant approach for enhancing AI-supported user experiences in fitness and social platforms such as SOULSTRIVENET.

Chapter 8: Conclusion and Future Work

8.1 Final Project Summary

This dissertation set out to design, develop, and evaluate a prototype AI-powered social fitness application named **SOULSTRIVENET**. The project combined a social feed concept, an AI-driven conversational fitness assistant, and personalised recommendation logic within a mobile application framework. Using a modular architecture built with Node.js, MongoDB, and React Native, the system demonstrated how artificial intelligence and prompt engineering can be effectively integrated into a social fitness context.

The completed prototype provides a functional proof-of-concept rather than a production-ready system. Nevertheless, it successfully demonstrates that AI-enhanced conversational interaction, when carefully designed and constrained, can improve user engagement, motivation, and perceived value within a fitness application. The project also illustrates how thoughtful architectural decisions

and iterative development can balance ambition with feasibility in a student-led environment.

8.2 Key Contributions

This work makes several important contributions at both a technical and conceptual level:

- It demonstrates the practical integration of a large language model–based conversational agent into a mobile social fitness application.
- It highlights the importance of prompt engineering as a primary mechanism for controlling AI behaviour, improving relevance, and reducing unsafe or hallucinated outputs.
- It presents a modular, extensible system architecture suitable for future expansion into real-time social interaction and advanced personalisation.
- It provides empirical and qualitative insights into how AI-driven conversational tone and contextual awareness influence user engagement in a fitness setting.

Collectively, these contributions show that even small-scale AI systems, when carefully engineered and evaluated, can deliver meaningful functionality and user value.

8.3 Future Work and Research Directions

While the project achieved its core objectives, it also revealed multiple opportunities for future development and research. One major direction is the implementation of real-time social features, including live feed updates, notifications, group challenges, and user-to-user messaging. These features would enable richer community dynamics and allow deeper evaluation of social interaction effects.

Further work should also focus on advancing machine-learning–based personalisation. By collecting longitudinal user data and feedback, more sophisticated recommendation techniques such as collaborative filtering, reinforcement learning, and adaptive workout planning could be implemented. Introducing persistent user profiles and long-term conversational memory would allow the AI assistant to provide more coherent, personalised, and evolving guidance over time.

Additional research is needed in the areas of safety, privacy, and ethics. Any real-world deployment would require robust data protection mechanisms, transparent consent models, moderation frameworks, and safeguards against harmful advice or over-dependence on AI systems. Large-scale, long-term user studies would also be necessary to evaluate behavioural, psychological, and health-related outcomes.

Finally, future iterations could explore integration with wearable devices and support for multi-modal content such as images and videos. These enhancements would enable richer data collection, improved personalisation, and stronger community engagement.

7.4 Final Conclusion

In conclusion, this dissertation demonstrates that it is feasible to design and implement an AI-enhanced social fitness platform within the scope of a student-led project. The SOULSTRIVENET prototype shows that combining conversational AI, prompt engineering, and user-centred design can significantly enhance engagement and motivation in a fitness application. Although the system remains at an early stage, the project provides a strong technical and conceptual foundation for future development. It also highlights the responsibilities associated with deploying AI in health-adjacent domains, emphasising the need for careful design, evaluation, and ethical consideration. Overall, this work contributes a practical proof-of-concept, critical insights, and a clear roadmap for future research into AI-driven social fitness platforms.

References

1.
Chew, H. S. J. (2022) 'The Use of Artificial Intelligence–Based Conversational Agents (Chatbots) for Weight Loss: Scoping Review and Practical Recommendations', *JMIR Medical Informatics*, 10(4). Available at: https://medinform.jmir.org/2022/4/e32578
(Accessed: 10 November 2025).

2
(Yoo Jung Oh, Jingwen Zhang, Min-Lin Fang & Yoshimi Fukuoka 2021) 'A systematic review of artificial intelligence chatbots for promoting physical activity, healthy diet, and weight loss', *International Journal of Behavioural Nutrition and Physical Activity*, 18:160. DOI: 10.1186/s12966-021-01224-6. Available at: https://ijbnpa.biomedcentral.com/articles/10.1186/s12966-021-01224-6
(Accessed: 12 November 2025).

3
Aggarwal, A., Tam, C.C., Wu, D. et al. (2023) 'Artificial Intelligence–Based Chatbots for Promoting Health Behavioural Changes: Systematic Review', *J Med Internet Res*, 25:e40789. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC10007007
(Accessed: 15 November 2025).

4
Jörke, M., Genç, D., Teutschbein, V. et al. (2025) ‘Bloom: Designing for LLM-Augmented Behaviour Change Interactions’, preprint. Available at ;https://arxiv.org/abs/2510.05449
(Accessed: 15 November 2025).

5
Zhu, J., Dallal, D.H., Gray, R.C. et al. (2021) ‘Personalization Paradox in Behaviour Change Apps: Lessons from a Social Comparison-Based Personalized App for Physical Activity’, preprint. Available at:https://arxiv.org/abs/2101.10020
(Accessed: 15 November 2025).

6
Quantellient Enterprises (2025) ‘Fitness — Key Features of AI-Powered Fitness Chatbots’. Available at:https://www.quantellient.com/posts/whitepapers/fitness
(Accessed: 16 November 2025).

7
KodyTechnolab (2025) ‘AI in Fitness Industry | The Future of Health and Exercise’. Available at: https://kodytechnolab.com/blog/ai-in-fitness-industry
(Accessed: 19 November 2025).

8
‘Algorithmic curation’ (2018) in *The International Encyclopaedia of Media Literacy*. Available at: https://en.wikipedia.org/wiki/Algorithmic_curation
(Accessed: 16 November 2025).

9
Hasan Beyariand Tareq Hashem The Role of Artificial Intelligence in Personalizing Social Media Marketing Strategies for Enhanced Customer Experience (18 May 2025). Available at: https://www.mdpi.com/2076-328X/15/5/700
(Accessed: 16 November 2025).

10
Dillys Larbi, Rolf Wynn, Marianne Vibeke Trondsen, Eirik Årsand, Paolo Zanaboni, Kerstin Denecke & Elia Gabarron (21 March 2024) ‘What do adults living with obesity want from a chatbot for physical activity? – a qualitative study’, Available at; https://link.springer.com/article/10.1186/s44247-024-

00070-3?fromPaywallRec=false
(Accessed: 16 November 2025).

11
Tahereh Saheb, Mouwafac Sidaoui, Bill Schmarzo(June 2024) 'Convergence of artificial intelligence with social media: A bibliometric & qualitative analysis' Available at;https://www.sciencedirect.com/science/article/pii/S277250302400032X#sec0002
(Accessed: 2 December 2025).

12
University of Wolverhampton (2018)
Guide to Writing a Literature Review.
Available at: https://www.wlv.ac.uk/lib/media/departments/lis/skills/study-guides/LS015-Guide-to-Writing-a-Literature-Review.pdf
(Accessed: 2 December 2025).

Bibliography

Websites / Documentation / APIs:

- Supabase (2025) *Realtime Databases for Modern Apps*. Available at: https://supabase.com/docs
  (Accessed: 1 December 2025).
- Git hub (2025) *Realtime Storage for Modern Apps*. Available at: https://github.com/
  (Accessed: 1 December 2025).

- React Native (2025) *React Native Documentation*. Available at: https://reactnative.dev/docs/getting-started
  (Accessed: 7 December 2025).
- Expo (2025) *Expo SDK Documentation*. Available at: https://docs.expo.dev/
  (Accessed: 1 December 2025).
- OpenRouter AI (2025) *OpenRouter API Documentation*. Available at: https://openrouter.ai/docs
  (Accessed: 1 December 2025).
- OpenAI (2025) *ChatGPT:* Available at: https://openai.com/chatgpt

(Accessed: 14 December 2025).
- Hugeicons(2025) Hugeicons: Available at ; https://hugeicons.com/icon/image-03-stroke-rounded (Accessed: 15 December 2025).
- PngTree(2025) PngTree :Available at :https://pngtree.com/so/avatar/2 (Accessed: 1 December 2025).

Tools / Libraries / Frameworks:

- Node.js (2025) *Node.js Documentation*. Available at: https://nodejs.org/en/docs/ (Accessed: 13 December 2025).
- React.js (2025) *React.js Official Documentation*. Available at: https://reactjs.org/docs/getting-started.html (Accessed: 13 December 2025).